

DEPARTMENT OF MATHEMATICS & STATISTICS

MTH 211: Theory of Statistics

(Session: 2025-26 Semester: II)

Problem Set 6

(Testing of Hypotheses – UMP test, MLR property)

1. Based on a random sample $X_1, \dots, X_n$, derive the UMP size- α test for testing $H_0: \theta \leq \theta_0$ against $H_1: \theta > \theta_0$ for the following populations:

(a) $f_\theta(x) = \begin{cases} \frac{1}{\theta} e^{-x/\theta}, & \text{if } x > 0, \theta > 0 \\ 0, & \text{otherwise.} \end{cases}$

(c) $f_\theta(x) = \begin{cases} \frac{\theta}{x^{\theta+1}}, & \text{if } x \geq 1, \theta > 0 \\ 0, & \text{otherwise.} \end{cases}$

(d) $f_\theta(x) = (\sqrt{2\pi}\theta)^{-1} e^{-x^2/2\theta^2}, -\infty < x < \infty, \theta > 0$

2. Derive the UMP size- α test for testing $H_0: \theta \geq \theta_0$ against $H_1: \theta < \theta_0$ based on a random sample $X_1, \dots, X_n$ from

$$f_\theta(x) = \begin{cases} \theta x^{\theta-1}, & \text{if } 0 < x < 1, \theta > 0 \\ 0, & \text{otherwise.} \end{cases}$$

3. Based on a single observation X from $Poisson(\theta)$, derive the UMP size- α test for testing $H_0: \theta \leq \theta_0$ against $H_1: \theta > \theta_0$

4. Verify whether the following families have MLR property in some statistic.

(a) $\mathcal{P} = \{f_\theta(x): -\infty < \theta < \infty\}; f_\theta(x) = \begin{cases} 1, & \text{if } \theta < x < \theta + 1, \\ 0, & \text{otherwise} \end{cases}$

(b) $\mathcal{P} = \{f_\theta(x): -\infty < \theta < \infty\}; f_\theta(x) = \begin{cases} \frac{1}{\pi} \frac{1}{1+(x-\theta)^2}, & \text{if } -\infty < x < \infty, \\ 0, & \text{otherwise} \end{cases}$

(c) $\mathcal{P} = \{f_\theta(x): 0 < \theta < \infty\}; f_\theta(x) = \begin{cases} \frac{1}{\theta} e^{-x/\theta}, & \text{if } x > 0, \\ 0, & \text{otherwise} \end{cases}$

5. Consider a random sample $X_1, \dots, X_n$ from $U(0, \theta), \theta > 0$. Based on the statistic in which the family has MLR property, if possible, construct at least two UMP size- α tests for $H_0: \theta \leq \theta_0$ against $H_1: \theta > \theta_0$.